

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS CS student (NC State, May 2026) with hands-on production experience building GenAI agents, RAG systems, and reasoning workflows. At Isomer AI, implemented MCP tool-calling across 8+ LLMs (Claude, GPT-4o, Gemini) in a production claims automation platform. Built Glance AI -- a compliance-ready RAG system with per-tenant data isolation for healthcare and fintech environments. Designed AlgoPulse, an event-driven AI orchestration platform handling 2,000 async inference requests/second with Temporal.io durable workflows. Experienced in PyTorch, TensorFlow, and the full applied ML lifecycle from pipeline to production deployment.

CORE COMPETENCIES

GenAI Agents & Reasoning Workflows

RAG Systems & Vector Retrieval

Multi-Model LLM Integration (MCP)

PyTorch & TensorFlow

Scalable Data Pipelines

Python (Applied ML)

Agentic Orchestration (Temporal.io)

Production ML Deployment

End-to-End Ownership

EXPERIENCE

Isomer AI May 2025 - Present
Product Engineer Intern - San Francisco, CA (Remote)

- Built **multi-model AI agent tooling** across Claude, GPT-4o, and Gemini using **MCP (Model Context Protocol)** - structured tool execution, function calling, and reasoning workflow orchestration across 8+ LLMs in production
- Shipped 10+ backend features across 7 deployments on TypeScript/Node.js/PostgreSQL/Redis; built real-time event pipelines, document processing (PDF/email/S3), and RBAC - **394 commits** to the production codebase
- Integrated Temporal.io for durable workflow orchestration across async AI reasoning loops; monitored deployed models with Sentry observability

NC State University Apr 2025 - Present
Graduate Research Assistant

- Applied **hyperparameter tuning and time-series ML** for genomic models, improving accuracy by 12%; built a cloud-based data platform (React + JBrowse 2) on GCP serving 100+ researchers across 500GB datasets

Mphasis (HP Company) Jun 2024 - Aug 2024
Software Developer Intern

- Built **5+ AI-driven workflows** including an OCR/NLP/ML health claim automation tool improving accuracy by 25%; automated model validation and red-teaming pipelines on AWS & GCP

Capgemini Jun 2023 - Aug 2023
AI/ML Intern

- Built **scalable data pipelines** with Kafka streaming + Spark batch processing for support ticket analysis; reduced latency by 20% and processing time by 15%

PROJECTS

Glance AI - Privacy-First RAG System Independent · 2025

Built a **compliance-ready RAG system** for healthcare and fintech environments with **per-tenant data isolation at the vector retrieval layer** -- 0 cross-tenant context leakage by design. Deterministic retrieval architecture eliminates hallucinations for regulated data environments.
Python · LLMs · Vector Databases · RAG · Compliance-ready architecture

AlgoPulse - Agentic AI Orchestration Platform Independent · 2025

Designed an event-driven AI orchestration platform, identifying a polling bottleneck and re-architecting to **Temporal.io durable reasoning workflows** + Kafka state sync + Supabase Edge Function streaming. Scaled to **2,000 async AI inference requests/second** with zero state loss under fault conditions.

TypeScript · Node.js · Temporal.io · Kafka · Supabase Edge Functions · React

PerCent - GenAI Finance App HackNC24 Top 10 of 100+ teams

RAG-based AI chatbot using Qwen-14B-Chat that reduced manual financial research by 30%. Built full-stack GenAI app with goal tracking, expense logging, and LLM-powered insights in 48 hours.

Python · Hugging Face · Qwen-14B-Chat · RAG · Selenium · React

Mapify - ML & NLP Pipeline IEEE Published

T5-transformer model pipeline that converted 1,500+ PDFs/images into structured mind maps and flowcharts, reducing academic review time by 60%. Presented to 500+ attendees at IEEE conference.

Python · T5 Transformer · PyTorch · NLTK · NetworkX

EDUCATION

North Carolina State University - MS Computer Science Aug 2024 - May 2026

Courses: Artificial Intelligence, Operating Systems, Data Science, Software Engineering, Algorithms

Mumbai University - BTech Computer Science Aug 2020 - May 2024

CGPA: 9.32/10 · Courses: DS & Algorithms, Machine Learning, Software Engineering, Big Data Infrastructure

SKILLS

AI/ML: PyTorch, TensorFlow, LangChain, MCP, RAG, Hugging Face, OpenAI API, Anthropic Claude, Google Gemini, Pandas, NumPy

Languages: Python, TypeScript, JavaScript, Java, SQL, C, R

Data Pipelines: Kafka, Spark, Hadoop, Supabase Edge Functions, Temporal.io, Redis

Cloud & Infra: AWS, GCP, Docker, Kubernetes, CI/CD, Grafana, Sentry

Web: Node.js, React, Next.js, Express, PostgreSQL, Git